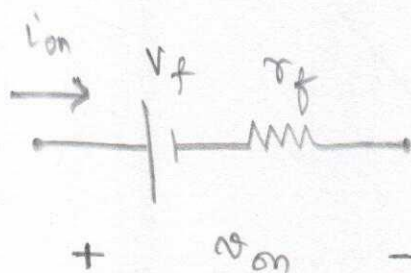


25.08.2025.

Assignment 1 Solutions.

1.



$$P_{avg} = \frac{1}{T} \int_0^T p_{on}(t) dt$$

$$= \frac{1}{T} \int_0^T v_{on}(t) \cdot i_{on}(t) dt$$

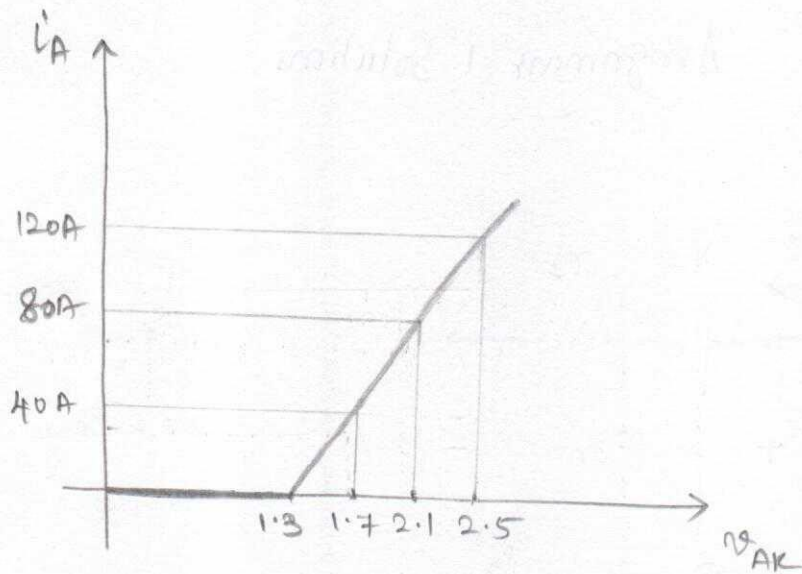
$$= \frac{1}{T} \int_0^T [V_f + i_{on}(t) \cdot r_f] i_{on}(t) dt$$

$$= V_f \cdot \left[\frac{1}{T} \int_0^T i_{on}(t) dt \right] + r_f \cdot \left[\frac{1}{T} \int_0^T i_{on}^2(t) dt \right]$$

$$\Rightarrow P_{avg} = V_f \cdot I_{on} + r_f \cdot I_{on,rms}^2$$

$$P_{avg} = V_f \cdot I_{on} + I_{on,rms}^2 \cdot r_f$$

2.



from the v - i characteristics,

$$V_f = 1.3V$$

$$r_f = \frac{1}{\text{slope of linear region of } v\text{-}i \text{ characteristics}}$$

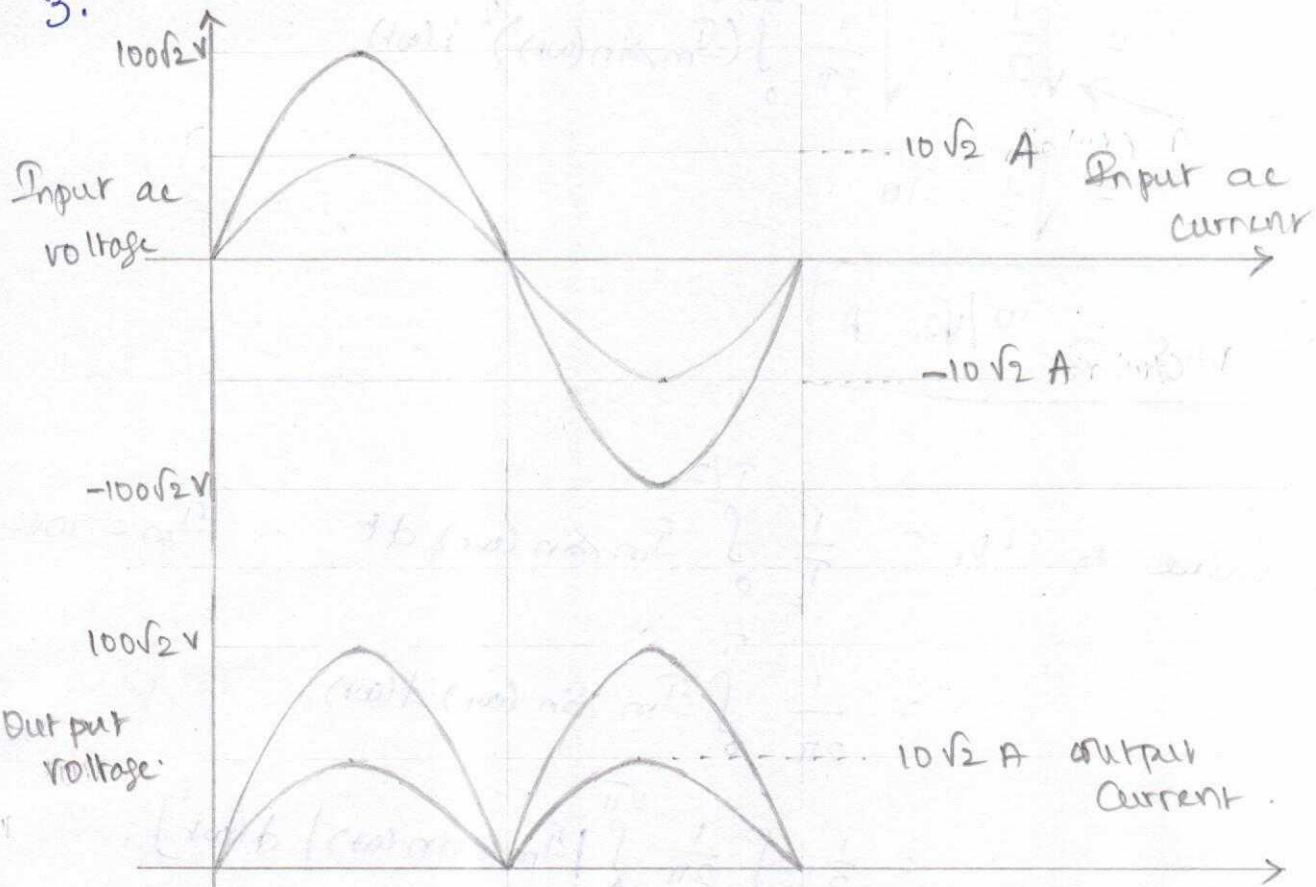
$$= \frac{\Delta v_{AK}}{\Delta i_A}$$

$$= \frac{2.1 - 1.7}{80 - 40}$$

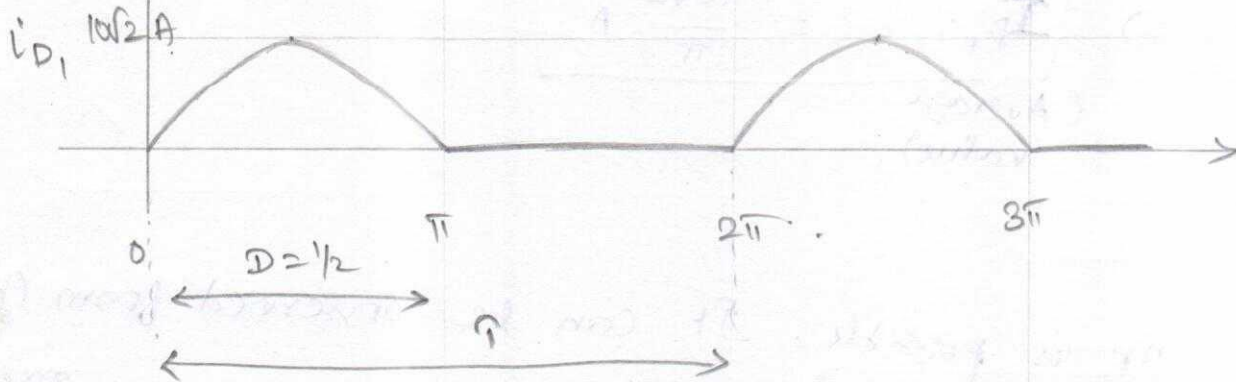
$$\Rightarrow r_f = 0.01 \Omega = 10 \text{ m}\Omega$$

$$V_f = 1.3V ; r_f = 10 \text{ m}\Omega$$

3.



a.



b.

$$\begin{aligned} \text{rms value of } i_{D1} &= \sqrt{\frac{1}{T} \int_0^{T/2} (I_m \sin \omega t)^2 dt} \quad I_m = 10\sqrt{2} A. \\ &= \sqrt{\frac{1}{2\pi} \int_0^{\pi} (I_m \sin \omega t)^2 d(\omega t)} \\ &= \sqrt{\frac{1}{2\pi} \left\{ \frac{1}{2} \left[\int_0^{\pi} (I_m \sin \omega t)^2 d(\omega t) \right] \right\}} \end{aligned}$$

both integral areas are the same.

$$\frac{1}{\sqrt{D}} \rightarrow \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (I_m \sin(\omega t))^2 d(\omega t)}$$

$$= \sqrt{\frac{1}{2}} \cdot 10$$

$$\Rightarrow \boxed{I_{D1, rms} = 10/\sqrt{2} \text{ A}}$$

Average value of $I_{D1} = \frac{1}{T} \int_0^{T/2} I_m \sin(\omega t) dt$. $I_m = 10\sqrt{2} \text{ A}$

$$= \frac{1}{2\pi} \int_0^{\pi} I_m \sin(\omega t) d(\omega t)$$

$$= \frac{1}{2} \left[\frac{1}{2\pi} \int_0^{2\pi} |I_m \sin(\omega t)| d(\omega t) \right]$$

$$\frac{1}{D} \rightarrow \frac{1}{2} \cdot \frac{2 \cdot I_m}{\pi}$$

$$= \frac{I_m}{\pi}$$

$$\Rightarrow \boxed{I_{D1} = \frac{10\sqrt{2}}{\pi} \text{ A}}$$

(Average value).

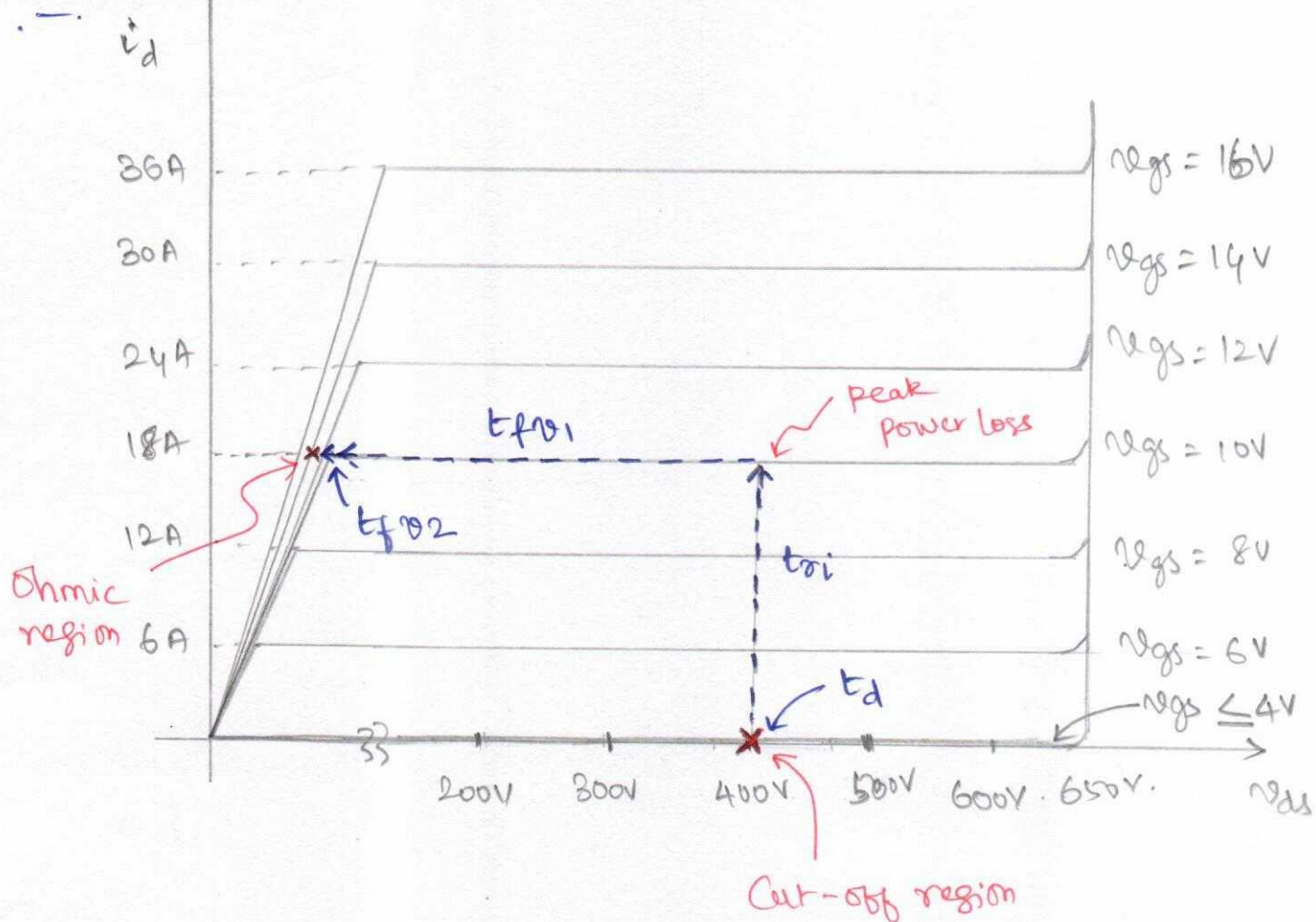
c. Yes. It is possible. It can be observed from (b)

That,

$$I_{D1, rms} = \frac{1}{\sqrt{D}} \times \text{rms value of rectified full sine wave} = \frac{1}{\sqrt{D}} \times \text{rms value of sine wave}$$

$$I_{D1} \text{ (Average value)} = \frac{1}{D} \times \text{average value of rectified full sine wave}$$

4 and 5.



6.

